

ALEKSANDR KOZLOV

Senior Technical Artist | Real-Time Optimization & Procedural Pipelines

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PROFILE

Senior Technical Artist with 10+ years in AAA production (PUBG Black Budget, Battlefield 2042, FIFA, King's Bounty II), specializing in bridging real-time engines with machine learning to solve production-scale problems. Deliver ML solutions for content optimization, combining a Deep Learning specialization (PyTorch, CNNs, Generative AI/CLIP) with deep Unreal Engine (C++), Houdini, and Python expertise to design data-driven tools and pipelines that scale content creation and performance optimization across teams.

CORE SKILLS

AI / Machine Learning: Python, PyTorch, Deep Learning, Convolutional Neural Networks, Generative AI (GANs, CLIP), ML-based asset & texture analysis

Engines & Languages: Unreal Engine 5 (C++, Blueprint), Houdini (VEX), C++, C#, Python

Pipelines & Automation: CI/CD (TeamCity, Docker), Procedural Content Generation, Build Automation

Performance & Profiling: PIX, Nvidia Nsight, Render Thread / GPU optimization

DCC & Tools: Blender, 3ds Max, Houdini, Photoshop, After Effects

Version Control: Perforce, Git, SVN

PROFESSIONAL EXPERIENCE

Senior Technical Artist / Tools Developer — PUBG Black Budget

Mar 2022 – Present

- **ML-driven optimization:** designed and shipped ML-based content analysis tooling (Python/PyTorch) to identify redundant assets and reduce memory footprint across the production pipeline.
- Applied deep learning models to content-optimization workflows, extending studio tooling from rule-based heuristics into data-driven asset analysis.
- Proposed and built an in-editor analytics tool surfacing DataLayer assignment and HLOD data for self-service performance visibility across the team.
- Proposed and delivered a foliage optimization strategy backed by a custom Houdini toolset, improving GPU and Render Thread performance.
- Implemented a custom HLOD system to raise visual fidelity without added performance cost; validated feasibility through targeted R&D and performance testing under a compressed timeline.
- Delivered automated optimization workflows using Unreal Engine commandlets, World Partition cell transformers, and Python scripts to reduce manual work and improve render performance at scale.
- Developed procedural tools for the environment department (biome scattering, spline tools, landscape editor) and Python editor tooling for Unreal Engine 5.
- Maintained recurring performance profiling (PIX, Nvidia Nsight) and reporting used across teams for early bottleneck detection.
- Authored technical documentation and led knowledge-sharing sessions to drive tool adoption across Art, Design, and QA departments.

Senior Technical Artist — Sperasoft

Aug 2021 – Mar 2022

Battlefield 2042, FIFA

- Built tools and scripts for Blender, Maya, and Frostbite; automated weapon-skin asset setup for weapon and environment artists.
- Supported artists with environment setup and troubleshooting within the Frostbite engine.

Technical Artist — 1C Entertainment

Nov 2019 – Jul 2021

King's Bounty II

- Built procedural Houdini tools and HDAs, including a snow/icicle generator, ivy tool, foliage scatter, cliff generator, and battle-zone generator.
- Automated texture-baking, UV-generation, and asset import/customization workflows with Python; developed Blueprints and UE4 shaders.

Technical Artist — VRTech Group

Nov 2017 – Nov 2019

- Created VFX for UE4 using Houdini and Niagara; built a procedural environment-generation system based on OSM and SRTM data.
- Implemented simulation systems (water flow, smoke, cloth dynamics) and optimized content for VR.

Motion Designer — NAOS

Jul 2016 – Oct 2017

- Developed visual effects (smoke, destruction, fire) and 3D motion graphics for project-mapping content; created VR showcases in Unreal Engine 4.

Lead 3D Artist / Motion Designer — Elephant Games

Jul 2011 – Jul 2016

- Led and supervised the 3D content department; animated cut-scenes in Adobe After Effects.
- Optimized team workflow via Python scripts (Blender) and JavaScript expressions (After Effects).

CERTIFICATIONS

- **Deep Learning Specialization** — Coursera (Apr–Dec 2022): Convolutional Neural Networks, Structuring Machine Learning Projects, Generative AI — from GANs to CLIP (Python & PyTorch), AI For Everyone.

EDUCATION

- Penza State University of Architecture and Construction — Specialist Degree, Urban Planning and Architecture (Sep 2007 – Jul 2011)